Quiz 13:-

(C11)

Converge or Diverge ?

(1) $\int_e^{\infty} \frac{1}{x \ln x}\, dx$

$= \int_e^{\infty} \left(\frac{1}{x}\right)\left(\frac{1}{\ln(x)}\right) dx$

$= \lim_{b\to\infty} \int_e^{b} \left(\frac{1}{x}\right)\left(\frac{1}{\ln(x)}\right) dx$

Let $u = \ln(x)$, $du = \frac{1}{x}\, dx$

$= \lim_{b\to\infty} \int_e^{b} \frac{1}{u}\, du$

$= \lim_{b\to\infty} \left[\ln|u| \right]_e^{b} = \lim_{b\to\infty} \ln\left|\ln(x)\right| \Big|_e^{b}$

$= \lim_{b\to\infty} \ln|b| - \ln|e| = \lim_{b\to\infty} \ln|b| - 0$

$= \infty$

$\therefore \int_e^{\infty} \frac{1}{x \ln(x)}\, dx$ is Diverge.

(B) $\int_1^e \frac{1}{x}\left(\frac{1}{\ln(x)}\right) dx$

Let $u = \ln(x)$, $du = \frac{1}{x}dx$

$= \int_1^e \frac{1}{u}\,du = \ln|u| \Big|_1^e$

$= \ln|\ln(x)| \Big|_1^e$

$= \ln|\ln(e)| - \ln|\ln(1)|$

$= 0 - \infty = -\infty$

$\therefore \int_1^e \frac{1}{x\ln(x)}\,dx$ is Diverge.